Cover Letter

19 August 2026

Editors of *Ledger*
University of Pittsburgh

Dear Editors,

I submit the manuscript **"MOSAIC: State-Transition-Centric Integrity Capsules for Distributed Ledger Execution"** for consideration as a full-length research article in *Ledger*.

The manuscript presents MOSAIC (Mutual-Obligation State And Integrity Capsules), an executable distributed-ledger protocol prototype organized around an authenticated state-transition lifecycle rather than treating a globally ordered block as the only integrity boundary. A Capsule names a predecessor StateSeal, operation, nonce, and witness obligations. Compatible WitnessReceipts and a ClosureProof authorize a deterministic successor root, while conflicting claims remain available as signed ConflictEvidence or an AbandonProof. The implementation connects this lifecycle to leaderless dissemination, durable SQLite WAL recovery, Ed25519 signatures, weighted membership and settlement, commit–reveal randomness, Reed–Solomon availability, deterministic execution, onboarding, and a hash-chained event log.

The paper is deliberately bounded in its claims. In a seven-process local rehearsal with two Byzantine test identities, a scheduled kill/restart, and partial-frame probes, all 120 requested operations succeeded. The measured liveness ratio was 1.0, unexpected operational errors were zero, median latency was 37.520503 ms, and p95 latency was 231.326768 ms. These measurements are explicitly identified as LOCAL_EMULATION results from one host. They are not presented as Internet-WAN measurements, proof of permissionless Sybil resistance, a production-mainnet claim, or evidence of universal superiority over HotStuff, Narwhal/Tusk, FastPay, Sui Lutris, Mysticeti, or any other ledger. The manuscript instead makes a reproducible protocol-composition claim and identifies the independent-WAN, formal, economic, and comparative experiments still required.

The work is relevant to *Ledger* because it addresses a general problem in cryptocurrency and distributed-ledger protocol design: how state transitions, conflicting claims, execution results, availability evidence, and recovery decisions can be represented as one auditable object lifecycle. The proposed design is not tied to one application, token, or vendor. The public repository contains the source code, tests, formal model, benchmark drivers, figures, tabular data, JSON artifacts, and event logs needed to inspect and reproduce the reported local experiment:

`https://github.com/adnanomar77/mosaic-protocol`

The repository is public and excludes private keys, credentials, and local virtual environments. The code is released under Apache-2.0; the manuscript and documentation are released under CC BY 4.0 where applicable.

An AI-assisted software and language workflow was used during development and manuscript preparation for code inspection, drafting support, organization, and language editing. The author reviewed the implementation, ran the experiments, checked the cited sources, verified the reported numbers, and remains solely responsible for the manuscript, its claims, its limitations, and the final submission. This use is disclosed in the manuscript in accordance with *Ledger*'s AI policy.

The manuscript is original, is not under consideration elsewhere, and has not been published in a peer-reviewed venue. No external funding was received. The author declares no known conflicts of interest relevant to this submission.

Suggested reviewers. The following scholars are suggested because their public institutional profiles document expertise directly relevant to distributed consensus, blockchain protocols, cryptographic security, and reproducible distributed systems. The author has no known personal, institutional, supervisory, employment, or recent co-authorship relationship with any of them. The editor should exclude any candidate for whom an undisclosed conflict or availability issue is identified; none has been contacted or has agreed to review.

1. **Christian Cachin** — Professor of Computer Science, University of Bern. Expertise: blockchain and consensus protocols, distributed computing, cryptographic protocols, and cloud-computing security. Public institutional email: `christian.cachin@unibe.ch`. Profile: `https://crypto.unibe.ch/cc/`

2. **Rachid Guerraoui** — Full Professor, Distributed Computing Laboratory, EPFL. Expertise: distributed algorithms, secure distributed storage, transactional shared memory, distributed programming languages, and

robust distributed computing. Public institutional email: `rachid.guerraoui@epfl.ch`. Profile: `https://people.epfl.ch/rachid.guerraoui?lang=en`

3. **Elaine Shi** — Professor of Computer Science and Electrical and Computer Engineering, Carnegie Mellon University. Expertise: blockchain, cryptography, distributed systems security, formal methods, protocol security, and verification. Public institutional email: `rshi@andrew.cmu.edu`. Profile: `https://www.cylab.cmu.edu/directory/bios/shi-elaine.html`

Thank you for considering this submission. I would welcome a rigorous review of both the protocol hypothesis and the limits of the current evidence.

Sincerely,

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